



Prospective Study

Evaluating thresholds of Mac-2 binding protein glycosylation isomer in association with clinical outcomes in patients with cirrhosis

Trung Hieu Doan, Khue Minh Nguyen, Xung Van Nguyen, Anh Thi Ngoc Pham, Nhan Duc Le

Specialty type: Gastroenterology and hepatology

Provenance and peer review: Unsolicited article; Externally peer reviewed.

Peer-review model: Single blind

Peer-review report's classification

Scientific Quality: Grade C

Novelty: Grade C

Creativity or Innovation: Grade C

Scientific Significance: Grade C

P-Reviewer: Zhang ZY, Assistant Professor, China

Received: May 8, 2025

Revised: June 28, 2025

Accepted: August 21, 2025

Published online: September 27, 2025

Processing time: 141 Days and 7.6 Hours



Trung Hieu Doan, Xung Van Nguyen, Department of Gastroenterology, Danang Hospital, Danang 50000, Viet Nam

Khue Minh Nguyen, Department of Genetics, Ho Chi Minh University of Science, Ho Chi Minh City 700000, Viet Nam

Khue Minh Nguyen, Department of Scientific Affairs, Sysmex Vietnam, Ho Chi Minh City 700000, Viet Nam

Anh Thi Ngoc Pham, Department of Laboratory, Danang Hospital, Danang 50000, Viet Nam

Nhan Duc Le, Intensive Care Unit, Danang Hospital, Danang 50000, Viet Nam

Co-first authors: Trung Hieu Doan and Khue Minh Nguyen.

Corresponding author: Nhan Duc Le, MD, PhD, Intensive Care Unit, Danang Hospital, 124 Hai Phong, Hai Chau, Danang 50000, Viet Nam. drnhandanang@gmail.com

Abstract

BACKGROUND

Cirrhosis is a progressive condition characterized by fibrosis that can lead to severe complications and increased mortality. The mac-2 binding protein glycosylation isomer (M2BPGi) is a prominent biomarker for predicting hepatocellular carcinoma (HCC) and cirrhosis-induced esophageal varices (EV).

AIM

To investigate thresholds of M2BPGi associated with HCC, EV, and decompensation in patients with cirrhosis.

METHODS

This was a prospective study. A total of 153 patients with cirrhosis who met the inclusion criteria were enrolled. The patients were diagnosed with HCC and EV according to the Baveno VII and European Association for the Study of the Liver guidelines. Baseline serum M2BPGi levels were assessed along with other routine tests. The data analysis aimed to determine the cutoff values of M2BPGi for predicting EV and HCC.

RESULTS

In the study 85.6% of patients were Child-Pugh B and C. M2BPGi mean cutoff

index was 7.1 ± 3.7 , showing no significant etiological differences. However, M2BPGi levels varied significantly among Child-Pugh classes, EV classifications, and between patients with and without HCC ($P < 0.01$). M2BPGi cutoff values for predicting HCC, EV, and decompensated cirrhosis were 6.50, 6.64, and 5.25, respectively. Multivariate analysis confirmed M2BPGi as an independent risk factor for EV [adjusted odds ratio (aOR): 1.3, 95%CI: 1.08-1.64] and liver decompensation (aOR: 2.11, 95%CI: 1.37-3.83). Area under the curve of M2BPGi for HCC differentiation was 0.71. An algorithm combining alpha-fetoprotein (AFP) and M2BPGi detected 26 of 28 HCC cases with 98.04% accuracy *vs* 10 cases by AFP alone.

CONCLUSION

Serum M2BPGi predicted cirrhosis complications, including decompensation and varices, especially in HCC. Combined with AFP, it enhanced HCC detection. Future liver biopsy studies are needed for tissue confirmation.

Key Words: Cirrhosis; Hepatocellular carcinoma; Esophageal varices; High-risk esophageal varices; Liver decompensation; Mac-2 binding protein glycosylation isomer

©The Author(s) 2025. Published by Baishideng Publishing Group Inc. All rights reserved.

Core Tip: This study introduces serum mac-2 binding protein glycosylation isomer (M2BPGi) as a novel biomarker for predicting complications in cirrhotic patients, particularly hepatocellular carcinoma (HCC), liver decompensation, and esophageal varices. The findings show that combining M2BPGi with alpha-fetoprotein significantly enhances diagnostic accuracy, achieving an impressive sensitivity of 92.8% for HCC detection. Importantly, M2BPGi levels correlate with disease severity as measured by the Child-Pugh classification, highlighting its potential role in clinical risk stratification.

Citation: Doan TH, Nguyen KM, Nguyen XV, Pham ATN, Le ND. Evaluating thresholds of Mac-2 binding protein glycosylation isomer in association with clinical outcomes in patients with cirrhosis. *World J Hepatol* 2025; 17(9): 109179

URL: <https://www.wjgnet.com/1948-5182/full/v17/i9/109179.htm>

DOI: <https://dx.doi.org/10.4254/wjh.v17.i9.109179>

INTRODUCTION

Cirrhosis represents an advanced stage in most chronic liver diseases. The 1-year mortality rate associated with cirrhosis exhibits significant variability, ranging from 1% to 57%, contingent on the degree of decompensation. Globally, it is the 14th leading cause of death and ranks 4th in Central Europe[1]. According to a Global Burden of Disease study conducted in 2020, cirrhosis accounted for approximately 1.32 million deaths in 2017, with about 0.44 million occurring in females and 0.88 million in males[2]. The mortality rate due to cirrhosis is particularly high in low-income regions such as sub-Saharan Africa, Central Asia, and Southeast Asia[2]. Notably, in Viet Nam, statistics from 2018 indicated that cirrhosis was the seventh leading cause of death among the top ten causes[3].

Cirrhosis can result in severe complications. Hepatocellular carcinoma (HCC), esophageal varices (EV), and liver decompensation are the most significant concerns in such patients. Indeed, 80%-90% of patients with HCC are diagnosed with cirrhosis[4]. Given the high prevalence of cirrhosis in patients with HCC, international guidelines strongly recommend identifying the risk of HCC development in such patients. Several markers have been developed and utilized for early detection to identify risk factors for HCC. These include alpha-fetoprotein (AFP), protein induced by vitamin K absence or antagonist-II, AFP-L3, liver stiffness assessed *via* ultrasound, and quantitative hepatitis B surface antigen in patients with hepatitis B infection. However, no optimal marker nor method has been established yet.

More specifically, previous studies demonstrated that the combination of AFP and ultrasound exhibited only 63% sensitivity for the early detection of HCC[5]. The limitation of the deployment of AFP in conjunction with ultrasound for HCC surveillance may lead to more than 40% of patients being misdiagnosed with early cases. Consequently, researchers are actively exploring new and more specific markers for the early detection of HCC in patients with chronic hepatitis, regardless of cirrhosis status. Protein induced by vitamin K absence or antagonist-II, AFP-L3, and particularly the GAAD score, have recently attracted the attention of numerous researchers and clinicians because the combination of individual parameters has improved HCC diagnostic performance[6]. Despite their superior attributes the high-cost individual parameters contributing to the GAAD score may constrain their usage among patients.

While effective treatment of the underlying causes of cirrhosis is important, identifying factors that can predict future complications, such as EV and liver decompensation, is crucial for triggering strict follow-up. Traditionally, an invasive method, esophagogastroduodenoscopy-retrieved parameters, consisting of size and the presence of red signs, was the gold standard for determining the existence of EV. Recent advancements in scientific and technical knowledge have led to the investigation of noninvasive methods aimed at ruling out the presence of EV in patients with cirrhosis, both in general and in specific contexts. These indicators, including platelet count and liver stiffness, accurately exclude low-risk EV in patients with cirrhosis[7]. Nevertheless, FibroScan has been found to be affected by obesity and acute inflammation, which may result in a false elevation. In addition, limited installation and well-served personnel requirements limit acc-

essibility in rural areas.

Mac-2 binding protein (M2BP) is a secreted glycoprotein (about 90 kDa) that contains seven N-glycans per monomer. M2BP exhibits high polymerization in serum, forming a large doughnut-like structure[8]. It is secreted by hepatic stellate cells (HSCs) in the liver and by macrophages from other organs. M2BP interacts with collagens IV, V, and VI, fibronectin, and nidogens[9]. Alterations in the glycan structure of M2BP reflect the degree of cell differentiation and carcinogenesis [10]. For instance, the N-glycosylation of M2BP changes during liver disease progression, with altered M2BP levels increasing in advanced fibrosis. Lectin *Wisteria floribunda* agglutinin specifically recognizes the altered M2BP produced by HSCs [M2BP glycosylation isomer (M2BPGi)][8]. HSCs are the primary producers of M2BPGi, which acts as a juxtacrine messenger enhancing mac-2 protein (galectin-3) expression in Kupffer cells. Mac-2 activates HSCs to become fibrogenic, reflecting HSC activation during fibrogenesis. Clinically, M2BPGi is recognized as a marker of fibrosis in various liver diseases[9].

Recent studies have highlighted the potential of M2BPGi as a biological marker for evaluating liver fibrosis. M2BPGi has been found to correlate with fibrosis stages in various etiologies, including viral hepatitis, metabolic dysfunction-associated steatotic liver disease, autoimmune hepatitis[11], and the prognosis of HCC in patients with chronic hepatitis B infection and chronic hepatitis C, regardless of treatment status[12,13]. Moreover, several studies have shown that M2BPGi is a promising short-term predictor of recurrent HCC in patients undergoing transarterial chemoembolization and hepatectomy[14]. Recently, M2BPGi has been found to be a potential marker to detect other cirrhosis-induced complications comprising EV, portal hypertension, and mortality[13,15].

Building on the existing literature that has explored the value of M2BPGi in liver fibrosis among patients with chronic hepatitis, this study further investigated its association with clinical outcomes in patients with cirrhosis. To address this concern we conducted a study titled “Assessing thresholds of M2BPGi in association with clinical outcomes in patients with cirrhosis”. This study aimed to investigate M2BPGi concentration in the association with HCC, EV, and decompensation in patients with all-cause cirrhosis. Additionally, we sought to improve diagnostic performance by combining M2BPGi with existing markers.

MATERIALS AND METHODS

Study design

This cross-sectional, prospective study was conducted at the Gastroenterology Department of Da Nang Hospital in Da Nang City, Viet Nam, from March 1, 2022 to September 31, 2024. A total of 153 patients with cirrhosis receiving treatment at the inpatient gastroenterology department were enrolled in the study and evaluated for HCC and EV. Blood sampling was performed on the same day as imaging diagnosis.

Patients

The diagnosis of cirrhosis was based on a comprehensive examination comprising the following: (1) Clinical signs and radiological findings, including morphological alterations in the liver (coarse liver echotexture with irregular margins), ascites, splenomegaly, and jaundice; (2) Laboratory test results: Aspartate aminotransferase (AST)/alanine aminotransferase (ALT) ratio > 1, total bilirubin > 17.1 $\mu\text{mol/L}$, serum albumin < 35 g/L, international normalized ratio > 1.3, or prolonged prothrombin time; and (3) FibroScan of ≥ 11 kPa and/or an AST to platelet ratio index ≥ 2 for patients with ambiguous clinical findings[16]. The exclusion criteria were as follows: (1) Acute decompensation; (2) End-stage decompensated cirrhosis and limited expected survival time; (3) Declined participation in the study; and (4) Under 18 years of age.

HCC was defined according to the 2018 American Association for the Study of Liver Diseases Guidelines: (1) Lesion > 1 cm and AFP > 20 ng/mL; and/or (2) Criteria on multiphase imaging by multiphase CT and magnetic resonance imaging[17]. The presence of EV was determined according to the Japan Gastroenterological Endoscopy Society classification: F0, no varicose appearance; F1, straight, small-caliber varices; F2, moderately enlarged, beady varices; and F3, markedly enlarged, nodular or tumor-shaped varices[18]. High-risk EVs were defined as medium or large varices, small varices with red signs, or Child–Pugh C in accordance with the European Association for the Study of Liver guidelines [19].

Routine laboratory tests

Blood samples were collected from all patients on the same day as the imaging diagnosis. Routine laboratory data were collected, including hemoglobin, WBC, platelet count, AST, ALT, albumin, total bilirubin, ammonia, and total bilirubin. AFP was measured by a Cobas C6000 analyzer (Roche Diagnostics, Indianapolis, IN, United States), and viral tests including HBV surface antigen and hepatitis C antibody were measured on an Architect i2000SR immunoassay analyzer (Abbott, Chicago, IL, United States). Prothrombin time was measured on a Sysmex coagulation analyzer (Sysmex Corp., Kobe, Hyogo, Japan). AST > 40 IU/L was defined as the upper limit. AST to platelet ratio index was calculated as: AST level (IU/L)/AST (upper limit of normal) (IU/L)/platelet level $\times 100$.

Liver elasticity measurement

Liver stiffness was measured using transient elastography (FibroScan; Echosens, Paris, France) by an experienced operator following established recommendations[20]. The results of at least 10 valid shots obtained in the session were expressed in kPa. A success rate of < 60% or an interquartile range/median value of > 30% was considered unreliable.

Computed tomography and magnetic resonance imaging

To confirm HCC cases all patients underwent computed tomography (CT) and/or magnetic resonance imaging (MRI) examinations on the same day as their clinical examination and blood sampling. A GE Optima 128-slice CT scanner (GE Healthcare, Chicago, IL, United States) and Philips 1.5T Multiva MRI system (Philips Healthcare, Best, Netherlands) with multiphase contrast enhancement were employed to detect HCC in patients with cirrhosis.

Esophagogastroduodenoscopy

Endoscopic procedures were conducted using the Olympus Evis X1, 170, and 190 systems (Olympus, Tokyo, Japan). The classification of variceal size was as follows: (1) Small, < 5 mm diameter; (2) Medium, 5-10 mm; and (3) Large, > 10 mm. Red signs were identified using narrow-band imaging technology, which enhances the contrast between blood vessels and surrounding tissues by employing different light wavelengths. Narrow-band imaging facilitates the visualization of red spots by increasing the visibility of vascular structures. Patients who did not exhibit these features were considered to be at low risk for variceal bleeding.

M2BPGi testing

Serum M2BPGi was measured in stored serum by employing a lectin-antibody sandwich immunoassay on an automated immunoassay system, HISCL-5000 (Sysmex), as previously reported[8]. Venous blood was collected prior to the endoscopy. Three milliliters of venous blood was centrifuged at 2000 rpm for 15 min at room temperature (25-30 °C) to obtain serum samples, which were then stored at -80 °C. The results were expressed as the cutoff index (C.O.I.).

Statistical analysis

Continuous variables were expressed as mean $\pm$ SD or median (interquartile range), and categorical data were reported as number (percentage). Differences between groups were evaluated using Student's *t*-test or the χ^2 statistic, as appropriate. The predictability of M2BPGi for complications, including liver decompensation, the presence of EV, high-risk EV, and HCC, was expressed as the area under the receiver operating characteristic curve (AUROC). Multivariate logistic regression was applied to exam diagnostic performance of the combination of various parameters in association with the presence of EV and liver decompensation. The optimal cutoff M2BPGi value was obtained by maximizing Youden's index, and statistical analysis was performed using GraphPad Prism v10 (GraphPad, La Jolla, CA, United States). Statistical significance was set at $P < 0.05$.

RESULTS

Baseline characteristics of patients

Table 1 shows the basic characteristics of the patients. The study population consisted predominantly of males (80.4%), with a mean age of 56.3 years. Liver function markers such as AST, ALT, and bilirubin levels indicated a significant degree of liver dysfunction, with notably elevated AST and bilirubin levels. Albumin levels decreased, consistent with impaired liver function. The platelet count also decreased, likely reflecting advanced liver disease or portal hypertension. A significant proportion of patients were classified as Child-Pugh class B (41.8%) or C (43.8%), indicating moderate-to-severe liver dysfunction. The mean FibroScan value of 36.4 kPa further corroborates the presence of significant liver fibrosis or cirrhosis. AFP levels are generally within a low-to-moderate range, while the M2BPGi marker also exhibited elevated values, which may indicate an underlying liver pathology.

Correlation between M2BPGi levels with severity of cirrhosis, model for end-stage liver disease score, and degree of EV and HCC

As shown in **Table 2**, serum M2BPGi levels significantly increased with the degree of cirrhosis (Child-Pugh) ($P < 0.01$), model for end-stage liver disease score ($P = 0.0018$), and severity of EV ($P < 0.01$). Notably in the group of patients with HCC, M2BPGi levels were also significantly higher than those in the group with cirrhosis without HCC ($P < 0.01$).

Association of M2BPGi with clinical outcomes in patients with cirrhosis

Analysis of the ROC curve was conducted to determine the cutoff value of M2BPGi levels for diagnosing HCC, liver decompensation, and the presence of EV (**Figure 1**). For HCC the ROC curve demonstrated an AUROC of 0.71 at the cutoff of 6.5 C.O.I., with sensitivity, specificity, negative predictive value (NPV), and positive predictive value (PPV) of 89.3%, 58.4%, 96.1%, and 32.5%, respectively (**Figure 1A**). Further investigation was performed using an algorithm combining M2BPGi (> 6.5 C.O.I.) and AFP (> 400 IU/L) to detect HCC cases. The algorithm resulted in a significant improvement in HCC detection with an accuracy of 98.0%. This algorithm, illustrated in **Figure 2**, yielded a sensitivity of 92.8%, a specificity of 99.2%, an NPV of 98.4%, and a PPV of 96.3%.

The results demonstrated that M2BPGi can distinguish liver decompensation at 5.25 C.O.I. with an AUROC of 0.69, balanced sensitivity (73.3%) and specificity (72.7%), and a high NPV of 94.1%, thus rendering it reliable for excluding decompensation (**Figure 1B**). Similarly, the model demonstrated moderate accuracy, with a specificity of 72.7% and a relatively low PPV of 30.8% for the presence of EV (cut-off: 6.64, AUROC: 0.68), indicating limited reliability in confirming positive cases (**Figure 1C**). For high-risk EV (cutoff: 6.86, AUROC: 0.63), the discriminatory power was weak, with fair sensitivity (60.7%) and specificity (65.3%), suggesting limited clinical utility (**Figure 1D**).

Table 1 Baseline characteristics of the study participants, *n* (%) / mean $\pm$ SD / median (interquartile range)

Characteristic, <i>n</i> = 153	Finding
Male	123 (80.4)
Age (years)	56.3 $\pm$ 10.9
AST (U/L)	93.2 (49.4-149.5)
ALT (U/L)	48.0 (30.2-82.9)
Total bilirubin (μ mol/L)	41.8 (20.7-69.8)
Albumin (g/L)	27.1 (22.0-33.1)
Hemoglobin (g/L)	111.0 (94.0-124.0)
Platelet as $\times 10^9$ /L	94.0 (59.0-145.0)
AFP (ng/mL)	4.66 (2.4-13.0)
Child-Pugh	
A	22 (14.4)
B	64 (41.8)
C	67 (43.8)
M2BPGi C.O.I.	7.1 $\pm$ 3.7
FibroScan in kPa, <i>n</i> = 77	36.4 $\pm$ 20.5

AST: Aspartate transaminase; ALT: Alanine transaminase; AFP: Alpha-fetoprotein; C.O.I.: Cutoff index; M2BPGi: Mac-2 binding protein glycosylation isomer.

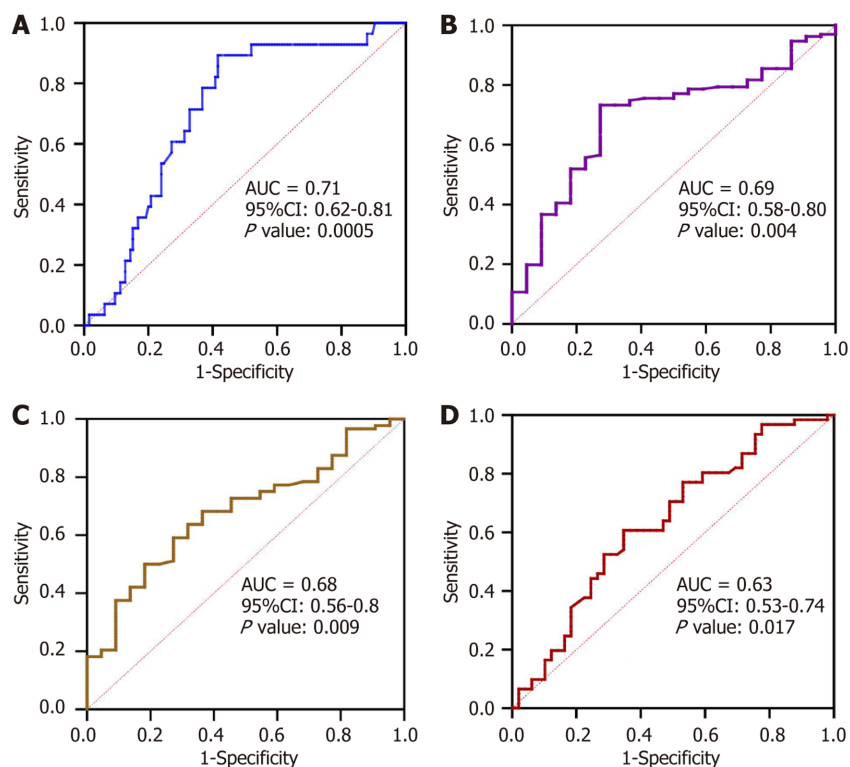


Figure 1 Area under the receiver operating characteristic curves of mac-2 binding protein glycosylation isomer in predicting hepatocellular carcinoma, liver decompensation, presence of esophageal varices, and high-risk esophageal varices. A: Hepatocellular carcinoma; B: Liver decompensation; C: Esophageal varices; D: High-risk esophageal varices. AUC: Area under the curve.

Table 2 Serum mac-2 binding protein glycosylation isomer according to the degree of cirrhosis, esophageal varices, and with or without hepatocellular carcinoma, mean $\pm$ SD

Characteristic	<i>n</i>	M2BPGi C.O.I.	<i>P</i> value ¹
Child-Pugh			
A	22	5.20 $\pm$ 2.65	< 0.001
B	64	6.14 $\pm$ 2.89	
C	67	8.63 $\pm$ 4.08	
MELD score			
≤ 9	38	6.09	0.003
10-19	53	6.43	
20-29	39	7.85	
30-39	10	10.02	
≥ 40	7	11.11	
Esophageal varices			
F0	22	5.3 $\pm$ 4.3	< 0.05
F1	10	5.0 $\pm$ 3.8	
FII	21	7.6 $\pm$ 3.4	
FIII	57	8.1 $\pm$ 3.8	
HCC			
Yes	28	8.8 $\pm$ 2.9	< 0.05
No	125	6.7 $\pm$ 3.8	

¹Analysis of variance was performed to evaluate the difference amongst the group; $P < 0.05$ was considered statistically significant.

HCC: Hepatocellular carcinoma; C.O.I.: Cutoff index; M2BPGi: Mac-2 binding protein glycosylation isomer; MELD: Model for end-stage liver disease.

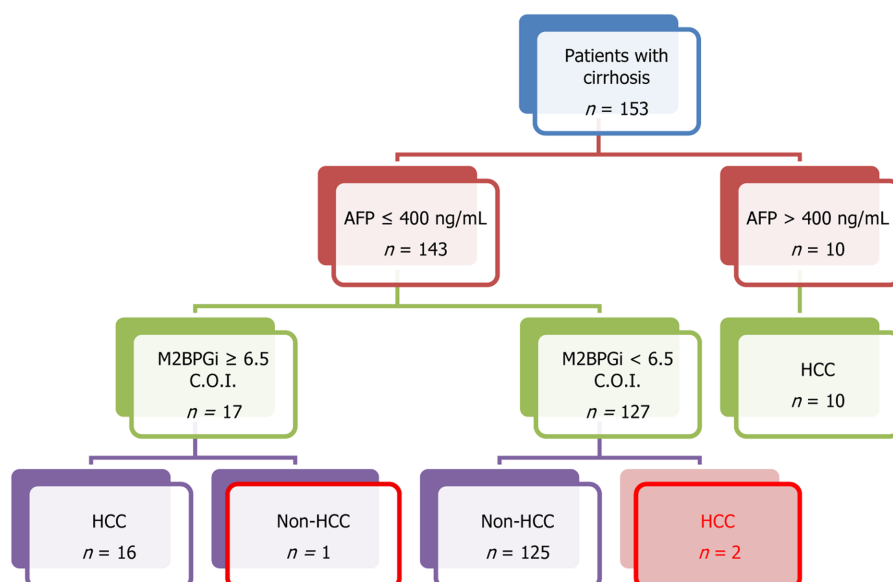


Figure 2 Algorithm combining alpha-fetoprotein and mac-2 binding protein glycosylation isomer to detect hepatocellular carcinoma in patients with cirrhosis. Hepatocellular carcinoma was confirmed by computed tomography or magnetic resonance imaging. AFP: Alpha-fetoprotein; C.O.I.: Cutoff index; M2BPGi: Mac-2 binding protein glycosylation isomer; HCC: Hepatocellular carcinoma.

As shown in Table 3, the multivariate logistic regression analysis revealed that M2BPGi was an independent risk factor associated with decompensation with an adjusted odds ratio (aOR) of 2.11, 95%CI: 1.37-3.83 and the presence of EV (aOR: 1.3, 95%CI: 1.08-1.64).

DISCUSSION

Our study demonstrated that M2BPGi is associated with HCC, liver compensation, and the presence of EV in patients with cirrhosis. M2BPGi has been identified as a fibrosis biomarker in patients with various etiologies including HBV, hepatitis C virus, metabolic dysfunction-associated steatotic liver disease, and autoimmune hepatology[8,13]. Furthermore, M2BPGi has been extensively investigated in East Asian populations as a biomarker for HCC[21]. Numerous studies have established the efficacy of M2BPGi in predicting HCC in patients with chronic hepatitis irrespective of the presence of cirrhosis[9,22]. In recent years, researchers have investigated the association between M2BPGi and EV and portal hypertension in patients with cirrhosis[11,23].

As shown in Table 2, serum M2BPGi levels significantly increased with cirrhosis severity (Child-Pugh score). These findings align with those of previous studies, such as that by Eso *et al*[24], who reported M2BPGi levels in Child-Pugh A, B, and C cirrhosis of 0.94, 4.775, and 11.37, respectively. Similarly, Hanai *et al*[25] observed serum Wisteria floribunda agglutinin positive-M2BP levels in Child-Pugh classes A, B, and C with C.O.I.s of 2.90, 6.15, and 9.45, respectively.

The present study demonstrated that an M2BPGi concentration of 6.5 C.O.I. can diagnose HCC in patients with cirrhosis, achieving an AUROC of 0.71 and a significant NPV of 96.1% ($P < 0.001$). These findings align with the research conducted by Liu *et al*[22] in which M2BPGi was identified as a substantial short-term predictor of HCC in chronic hepatitis B patients, particularly within 1 year to 2 years before diagnosis, with AUROCs exceeding 0.80. Tseng *et al*[26] reported that elevated levels of M2BPGi independently predicted HCC risk and enhanced existing models such as PAGE-B. Moreover, Jun *et al*[21] demonstrated that M2BPGi outperformed AFP in predicting HCC risk, particularly in patients with HBV without advanced fibrosis. Su *et al*[9] noted that M2BPGi levels declined significantly in patients without HCC but remained elevated in HCC cases. These findings further support the use of M2BPGi as a dynamic surveillance tool.

This study also revealed that M2BPGi levels of ≥ 6.5 C.O.I. considerably increased the risk of HCC, underscoring its clinical significance for early detection. Despite the high NPV for ruling out HCC, M2BPGi exhibited an inferior PPV. While M2BPGi demonstrated high sensitivity and low specificity, AFP showed a contrasting pattern (data not shown). To enhance the diagnosis of HCC, the algorithm combining M2BPGi and AFP showed superior performance compared to either individual M2BPGi or AFP. Specifically, this algorithm significantly improved the sensitivity, specificity, and PPV for diagnosing HCC and mitigated the limitations of these assays. Indeed, 16 out of 28 HCC cases with AFP levels lower than 400 ng/L were greater than 6.5 C.O.I. Collectively, this algorithm suggests a novel approach combining these assays to improve the early detection of HCC and reduce misdiagnoses caused by the low sensitivity of AFP. However, this model requires validation in a larger population, owing to the small sample size.

Regarding liver decompensation, our study identified a C.O.I. of 5.25 for M2BPGi in predicting liver decompensation, yielding an AUROC of 0.69, sensitivity of 73.3%, and specificity of 72.7%. Multivariate regression analysis demonstrated that M2BPGi was an independent risk factor for liver decompensation (aOR: 2.11; 95%CI: 1.37-3.83). Previous studies have indicated that elevated M2BPGi levels are associated with worsening of fibrosis and decompensation. For instance, M2BPGi was identified as an independent predictor of clinical decompensation in patients with compensated cirrhosis with an OR of 11.958 (95%CI: 1.876-76.226, $P = 0.009$)[27]. Uojima *et al*[28] observed that patients with decompensated cirrhosis exhibited significantly higher M2BPGi levels (6.91 ± 5.04) than patients with compensated cirrhosis (2.22 ± 1.61 , $P < 0.0001$), with a C.O.I. of 3.37 predicting decompensated cirrhosis with 77.8% sensitivity and 86.7% specificity. These findings demonstrated a relationship between M2BPGi levels and liver decompensation progression, although a universal cutoff value is lacking, necessitating larger studies for standardization.

The final clinical outcome examined in this study was the presence of EV. The M2BPGi demonstrated a fair discriminative ability. The sensitivity was moderate (59.1%), while the specificity was higher (72.7%) for identifying patients without EV. The PPV was 89.7%, indicating a strong correlation between positive test results and the presence of EV. However, its low NPV of 30.8% limits its efficacy in ruling out EV. Wu *et al*[29] demonstrated a positive correlation between M2BPGi and hepatic venous pressure gradient (HVPG) in patients with cirrhosis, reporting M2BPGi levels of 2.33 C.O.I. for clinically significant portal hypertension (HVPG > 10 mmHg) and 5.93 C.O.I. for severe portal hypertension (HVPG > 16 mmHg).

Kikukawa *et al*[15] also reported higher M2BPGi levels in patients with esophagogastric varices compared with those without esophagogastric varices. Furthermore, this study indicated that the M2BPGi value had predictive ability for high-risk EV, with an AUROC of 0.63 ($P < 0.05$). A previous study found higher M2BPGi levels in patients with high-risk EV than in those without (C.O.I. of 11.4 *vs* 3.7, $P < 0.001$), using a C.O.I. of 5; the sensitivity, specificity, PPV, and NPV were 92.6%, 70.1%, 55.6%, and 95.9%, respectively[23]. Our study demonstrated lower sensitivity and NPV (60.7% and 56.8%, respectively), which may indicate the limitation of this test in excluding patients with high-risk EV. Another factor contributing to this result was the sample size of the subset.

This study had several limitations. First, as this was a single-center study with a small sample size, the findings may not be generalizable to the entire target population. Second, the use of FibroScan for staging fibrosis may introduce bias owing to its inherent limitations. Additionally, the lack of liver histology deferred us from HCC confirmation. Hence, MRI/CT was used as a reference method due to its non-invasive nature. Third, given the cross-sectional nature of this study and the collection of data solely at baseline, the absence of longitudinal measurements constrained our ability to observe temporal changes in M2BPGi. Fourth, our algorithm combining AFP and M2BPGi was investigated in a small

Table 3 Univariate logistic regression and multivariate logistic regression in prediction of liver decompensation and esophageal varices

Variable	Univariate analysis		Multivariate analysis	
	OR	95%CI	aOR	95%CI
Liver decompensation				
MELD score	1.19	1.10-1.32	1.18	1.04-1.41
M2BPGi	1.23	1.06-1.47	2.11	1.37-3.83
FibroScan	1.06	1.02-1.12	1.05	0.98-1.15
AAR	1.80	1.06-3.48	4.12	0.84-35.8
Esophageal varices				
Platelets	0.99	0.99-1.00	1.00	0.99-1.02
FibroScan	0.99	0.97-1.02	0.98	0.95-1.00
M2BPGi	1.10	1.01-1.22	1.30	1.08-1.64
AAR	1.02	0.78-1.36	1.04	0.62-1.69
Child-Pugh				
A	Reference		Reference	
B	4.75	1.70-14.80	4.65	1.06-24.11
C	4.21	1.52-13.01	1.90	0.32-11.90

AAR: Aspartate aminotransferase to alanine aminotransferase ratio; aOR: Adjusted odds ratio; M2BPGi: Mac-2 binding protein glycosylation isomer; MELD: Model for end-stage liver disease; OR: Odds ratio.

sample size as HCC cases diagnosed in our patient population may have resulted in biased data. Thus, we suggest a larger population to investigate this algorithm in an establishment and validation cohort before its implementation in practical procedures. Furthermore, the predominance of males and treatment history may contribute to biased cutoffs. Finally, the limitation of the subset sample sizes (*e.g.*, EV, high-risk EV, and liver decompensation) precluded comprehensive data analysis. Future prospective multicenter studies should address and uncover the barriers. Nevertheless, the study results may provide a novel approach for diagnosing the aforementioned cirrhosis-induced complications in limited-resource settings.

CONCLUSION

Serum M2BPGi appears to be a valuable and accessible non-invasive biomarker for predicting clinically significant complications in patients with cirrhosis, such as hepatic decompensation and EV, and especially in cases with HCC. Although histological validation was not performed, the findings provided supportive clinical evidence for the utility of M2BPGi in risk stratification. Furthermore, the combined use of M2BPGi and AFP showed improved sensitivity in identifying HCC, highlighting its potential value in enhancing surveillance strategies in cirrhotic patients. Future studies incorporating liver biopsy are warranted to further confirm these associations exist at the tissue level.

FOOTNOTES

Author contributions: Doan TH and Nguyen KM contributed to the design of the study, revised the collected data, performed the statistical data analysis and interpretation, and reviewed and provided the final version of the manuscript; Nguyen XV and Pham ATN shared the operational work of the study and the writing of the manuscript; Doan TH and Nguyen XV recruited participants and collected data; Le ND supervised the study; Doan TH and Nguyen KM contributed equally to this study and merit co-first authorship; all authors contributed to revision of the manuscript for important intellectual content and approved the final version.

Institutional review board statement: The study was reviewed and approved by the Institutional Review Board of the Da Nang Hospital (No. 2022.11.54 issued in March 2022).

Informed consent statement: Informed consent was waived for this study based upon the utilization of residual blood samples.

Conflict-of-interest statement: Khue Minh Nguyen is a Sysmex employee. All other authors declare no conflicts of interest.

Data sharing statement: Technical appendix, statistical code, and dataset available from the co-first authors at minhkhueguyen8888@gmail.com or drdoanhieutrongbvdn@gmail.com. No additional data are available.

CONSORT 2010 statement: The authors have read the CONSORT 2010 statement, and the manuscript was prepared and revised according to the CONSORT 2010 statement.

Open Access: This article is an open-access article that was selected by an in-house editor and fully peer-reviewed by external reviewers. It is distributed in accordance with the Creative Commons Attribution NonCommercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited and the use is non-commercial. See: <https://creativecommons.org/licenses/by-nc/4.0/>

Country of origin: Viet Nam

ORCID number: Khue Minh Nguyen [0009-0006-0992-4092](https://orcid.org/0009-0006-0992-4092).

Corresponding Author's Membership in Professional Societies: Vietnam Association of Gastroenterology.

S-Editor: Liu H

L-Editor: A

P-Editor: Zhang YL

REFERENCES

- Gu W, Hortlik H, Erasmus HP, Schaaf L, Zeleke Y, Uschner FE, Ferstl P, Schulz M, Peiffer KH, Queck A, Sauerbruch T, Brol MJ, Rohde G, Sanchez C, Moreau R, Arroyo V, Zeuzem S, Welsch C, Trebicka J. Trends and the course of liver cirrhosis and its complications in Germany: Nationwide population-based study (2005 to 2018). *Lancet Reg Health Eur* 2022; **12**: 100240 [RCA] [PMID: 34901909 DOI: [10.1016/j.lanepe.2021.100240](https://doi.org/10.1016/j.lanepe.2021.100240)] [FullText] [Full Text(PDF)]
- GBD 2017 Cirrhosis Collaborators. The global, regional, and national burden of cirrhosis by cause in 195 countries and territories, 1990-2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet Gastroenterol Hepatol* 2020; **5**: 245-266 [RCA] [PMID: 31981519 DOI: [10.1016/S2468-1253\(19\)30349-8](https://doi.org/10.1016/S2468-1253(19)30349-8)] [FullText] [Full Text(PDF)]
- Ta H, Lin B, Palaniappan L. Vietnamese and Vietnamese-American health statistics, 2003-2019. *Journal of Asian Health*. 2020. [cited 12 August, 2025]. Available from: <https://med.stanford.edu/content/dam/sm/care/VNDataBrief.pdf>
- Simonetti RG, Cammà C, Fiorello F, Politi F, D'Amico G, Pagliaro L. Hepatocellular carcinoma. A worldwide problem and the major risk factors. *Dig Dis Sci* 1991; **36**: 962-972 [RCA] [PMID: 1649041 DOI: [10.1007/BF01297149](https://doi.org/10.1007/BF01297149)] [FullText]
- Tzartzeva K, Obi J, Rich NE, Parikh ND, Marrero JA, Yopp A, Waljee AK, Singal AG. Surveillance Imaging and Alpha Fetoprotein for Early Detection of Hepatocellular Carcinoma in Patients With Cirrhosis: A Meta-analysis. *Gastroenterology* 2018; **154**: 1706-1718.e1 [RCA] [PMID: 29425931 DOI: [10.1053/j.gastro.2018.01.064](https://doi.org/10.1053/j.gastro.2018.01.064)] [FullText]
- Piratvisuth T, Hou J, Tanwandee T, Berg T, Vogel A, Trojan J, De Toni EN, Kudo M, Eiblmaier A, Klein HG, Hegel JK, Madin K, Kroeniger K, Sharma A, Chan HLY. Development and clinical validation of a novel algorithmic score (GAAD) for detecting HCC in prospective cohort studies. *Hepatal Commun* 2023; **7**: e0317 [RCA] [PMID: 37938100 DOI: [10.1097/HC9.0000000000000317](https://doi.org/10.1097/HC9.0000000000000317)] [FullText] [Full Text(PDF)]
- Ding NS, Nguyen T, Iser DM, Hong T, Flanagan E, Wong A, Luiz L, Tan JY, Fulforth J, Holmes J, Ryan M, Bell SJ, Desmond PV, Roberts SK, Lubel J, Kemp W, Thompson AJ. Liver stiffness plus platelet count can be used to exclude high-risk oesophageal varices. *Liver Int* 2016; **36**: 240-245 [RCA] [PMID: 26212020 DOI: [10.1111/liv.12916](https://doi.org/10.1111/liv.12916)] [FullText]
- Kuno A, Ikehara Y, Tanaka Y, Ito K, Matsuda A, Sekiya S, Hige S, Sakamoto M, Kage M, Mizokami M, Narimatsu H. A serum "sweet-doughnut" protein facilitates fibrosis evaluation and therapy assessment in patients with viral hepatitis. *Sci Rep* 2013; **3**: 1065 [RCA] [PMID: 23323209 DOI: [10.1038/srep01065](https://doi.org/10.1038/srep01065)] [FullText] [Full Text(PDF)]
- Su TH, Peng CY, Tseng TC, Yang HC, Liu CJ, Liu CH, Chen PJ, Chen DS, Kao JH. Serum Mac-2-Binding Protein Glycosylation Isomer at Virological Remission Predicts Hepatocellular Carcinoma and Death in Chronic Hepatitis B-Related Cirrhosis. *J Infect Dis* 2020; **221**: 589-597 [RCA] [PMID: 31574141 DOI: [10.1093/infdis/jiz496](https://doi.org/10.1093/infdis/jiz496)] [FullText]
- Narimatsu H. Development of M2BPGi: a novel fibrosis serum glyco-biomarker for chronic hepatitis/cirrhosis diagnostics. *Expert Rev Proteomics* 2015; **12**: 683-693 [RCA] [PMID: 26394846 DOI: [10.1586/14789450.2015.1084874](https://doi.org/10.1586/14789450.2015.1084874)] [FullText]
- Shirabe K, Bekki Y, Gantumur D, Araki K, Ishii N, Kuno A, Narimatsu H, Mizokami M. Mac-2 binding protein glycan isomer (M2BPGi) is a new serum biomarker for assessing liver fibrosis: more than a biomarker of liver fibrosis. *J Gastroenterol* 2018; **53**: 819-826 [RCA] [PMID: 29318378 DOI: [10.1007/s00535-017-1425-z](https://doi.org/10.1007/s00535-017-1425-z)] [FullText]
- Bui HH, Nguyen ST, Phan ST, Nguyen KM, Nguyen CD. Evaluating M2BPGi as a Marker for Liver Fibrosis in Patients with Chronic Hepatitis B. *Dig Dis Sci* 2023; **68**: 4407-4417 [RCA] [PMID: 37861877 DOI: [10.1007/s10620-023-08143-5](https://doi.org/10.1007/s10620-023-08143-5)] [FullText] [Full Text(PDF)]
- Tamaki N, Kurosaki M, Loomba R, Izumi N. Clinical Utility of Mac-2 Binding Protein Glycosylation Isomer in Chronic Liver Diseases. *Ann Lab Med* 2021; **41**: 16-24 [RCA] [PMID: 32829576 DOI: [10.3343/alm.2021.41.1.16](https://doi.org/10.3343/alm.2021.41.1.16)] [FullText] [Full Text(PDF)]
- Lee IC, Lei HJ, Wang LC, Yeh YC, Chau GY, Hsia CY, Chou SC, Luo JC, Hou MC, Huang YH. M2BPGi Correlated with Immunological Biomarkers and Further Stratified Recurrence Risk in Patients with Hepatocellular Carcinoma. *Liver Cancer* 2025; **14**: 68-79 [RCA] [PMID: 40144467 DOI: [10.1159/000540802](https://doi.org/10.1159/000540802)] [FullText] [Full Text(PDF)]
- Kikukawa K, Uchida-Kobayashi S, Tamori A, Yoshida K, Kotani K, Motoyama H, Kozuka R, Hagihara A, Fujii H, Morikawa H, Enomoto M, Murakami Y, Kawada N. Serum Mac-2-binding protein glycosylation isomer predicts esophagogastric varices in cirrhotic patients with chronic hepatitis C virus infection treated with IFN-free direct-acting antiviral agent: M2BPGi levels predict varices in SVR patients. *Ann Hepatol* 2020; **19**: 367-372 [RCA] [PMID: 32444247 DOI: [10.1016/j.aohp.2020.04.002](https://doi.org/10.1016/j.aohp.2020.04.002)] [FullText]

- 16 **World Health Organization.** Guidelines for the prevention care and treatment of persons with chronic hepatitis B infection. 2015. [cited 12 August, 2025]. Available from: <https://www.who.int/publications/i/item/9789241549059>
- 17 **Marrero JA,** Kulik LM, Sirlin CB, Zhu AX, Finn RS, Abecassis MM, Roberts LR, Heimbach JK. Diagnosis, Staging, and Management of Hepatocellular Carcinoma: 2018 Practice Guidance by the American Association for the Study of Liver Diseases. *Hepatology* 2018; **68**: 723-750 [RCA] [PMID: 29624699 DOI: 10.1002/hep.29913] [FullText]
- 18 **Tajiri T,** Yoshida H, Obara K, Onji M, Kage M, Kitano S, Kokudo N, Kokubu S, Sakaida I, Sata M, Tajiri H, Tsukada K, Nonami T, Hashizume M, Hirota S, Murashima N, Moriyasu F, Saigenji K, Makuuchi H, Oho K, Yoshida T, Suzuki H, Hasumi A, Okita K, Futagawa S, Idezuki Y. General rules for recording endoscopic findings of esophagogastric varices (2nd edition). *Dig Endosc* 2010; **22**: 1-9 [RCA] [PMID: 20078657 DOI: 10.1111/j.1443-1661.2009.00929.x] [FullText]
- 19 **European Association for the Study of the Liver.** EASL Clinical Practice Guidelines for the management of patients with decompensated cirrhosis. *J Hepatol* 2018; **69**: 406-460 [RCA] [PMID: 29653741 DOI: 10.1016/j.jhep.2018.03.024] [FullText]
- 20 **Shiina T,** Nightingale KR, Palmeri ML, Hall TJ, Bamber JC, Barr RG, Castera L, Choi BI, Chou YH, Cosgrove D, Dietrich CF, Ding H, Amy D, Farrokh A, Ferraioli G, Filice C, Friedrich-Rust M, Nakashima K, Schafer F, Sporea I, Suzuki S, Wilson S, Kudo M. WFUMB guidelines and recommendations for clinical use of ultrasound elastography: Part 1: basic principles and terminology. *Ultrasound Med Biol* 2015; **41**: 1126-1147 [RCA] [PMID: 25805059 DOI: 10.1016/j.ultrasmedbio.2015.03.009] [FullText]
- 21 **Jun T,** Hsu YC, Ogawa S, Huang YT, Yeh ML, Tseng CH, Huang CF, Tai CM, Dai CY, Huang JF, Chuang WL, Yu ML, Tanaka Y, Nguyen MH. Mac-2 Binding Protein Glycosylation Isomer as a Hepatocellular Carcinoma Marker in Patients With Chronic Hepatitis B or C Infection. *Hepatol Commun* 2019; **3**: 493-503 [RCA] [PMID: 30976740 DOI: 10.1002/hep4.1321] [FullText] [Full Text(PDF)]
- 22 **Liu J,** Hu HH, Lee MH, Korenaga M, Jen CL, Batrla-Utermann R, Lu SN, Wang LY, Mizokami M, Chen CJ, Yang HI. Serum Levels of M2BPGi as Short-Term Predictors of Hepatocellular Carcinoma in Untreated Chronic Hepatitis B Patients. *Sci Rep* 2017; **7**: 14352 [RCA] [PMID: 29085039 DOI: 10.1038/s41598-017-14747-5] [FullText] [Full Text(PDF)]
- 23 **Nababan SHH,** Kalista KF, Jasirwan CO, Kurniawan J, Lesmana CRA, Sulaiman AS, Hasan I, Gani RA. Mac-2 Binding Protein Glycosylation Isomer for Screening High-Risk Esophageal Varices in Liver Cirrhotic Patient. *Livers* 2021; **1**: 60-67 [RCA] [DOI: 10.3390/livers1020006] [FullText]
- 24 **Eso Y,** Takai A, Taura K, Takahashi K, Ueda Y, Marusawa H, Seno H. Association of Mac-2-binding protein glycosylation isomer level with nutritional status in chronic liver disease. *J Gastroenterol Hepatol* 2018 [RCA] [PMID: 29473206 DOI: 10.1111/jgh.14130] [FullText]
- 25 **Hanai T,** Shiraki M, Ohnishi S, Miyazaki T, Ideta T, Kochi T, Imai K, Suetsugu A, Takai K, Shimizu M, Moriaki H. Impact of serum glycosylated Wisteria floribunda agglutinin positive Mac-2 binding protein levels on liver functional reserves and mortality in patients with liver cirrhosis. *Hepatol Res* 2015; **45**: 1083-1090 [RCA] [PMID: 25565570 DOI: 10.1111/hepr.12473] [FullText]
- 26 **Tseng TC,** Peng CY, Hsu YC, Su TH, Wang CC, Liu CJ, Yang HC, Yang WT, Lin CH, Yu ML, Lai HC, Tanaka Y, Nguyen MH, Liu CH, Chen PJ, Chen DS, Kao JH. Baseline Mac-2 Binding Protein Glycosylation Isomer Level Stratifies Risks of Hepatocellular Carcinoma in Chronic Hepatitis B Patients with Oral Antiviral Therapy. *Liver Cancer* 2020; **9**: 207-220 [RCA] [PMID: 32399434 DOI: 10.1159/000504650] [FullText]
- 27 **Xu WP,** Wang ZR, Zou X, Zhao C, Wang R, Shi PM, Yuan ZL, Yang F, Zeng X, Wang PQ, Sultan S, Zhang Y, Xie WF. Serum Wisteria floribunda agglutinin-positive Mac-2-binding protein evaluates liver function and predicts prognosis in liver cirrhosis. *J Dig Dis* 2018; **19**: 242-253 [RCA] [PMID: 29607614 DOI: 10.1111/1751-2980.12596] [FullText]
- 28 **Uojima H,** Hidaka H, Tanaka Y, Inoue T, Onoue M, Wada N, Kubota K, Nakazawa T, Shibuya A, Fujikawa T, Nakayama T, Yamanoue H, Sung JH, Kako M, Koizumi W. Wisteria floribunda agglutinin-positive human Mac-2 binding protein in decompensated cirrhosis. *J Gastroenterol Hepatol* 2018; **33**: 1889-1896 [RCA] [PMID: 29737582 DOI: 10.1111/jgh.14277] [FullText]
- 29 **Wu PS,** Hsieh YC, Lee KC, Huang YH, Hou MC, Lin HC. Mac-2 binding protein glycosylation isomer is a potential biomarker to predict portal hypertension and bacterial infection in cirrhotic patients. *PLoS One* 2021; **16**: e0258589 [RCA] [PMID: 34648567 DOI: 10.1371/journal.pone.0258589] [FullText] [Full Text(PDF)]



Published by **Baishideng Publishing Group Inc**
7041 Koll Center Parkway, Suite 160, Pleasanton, CA 94566, USA

Telephone: +1-925-3991568

E-mail: office@baishideng.com

Help Desk: <https://www.f6publishing.com/helpdesk>

<https://www.wjgnet.com>

